



Evaluating the impact of
immersive technologies on
STEM EDUCATION
in Sri Lanka's secondary
and tertiary curriculum



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INTRO



RESEARCH QUESTION

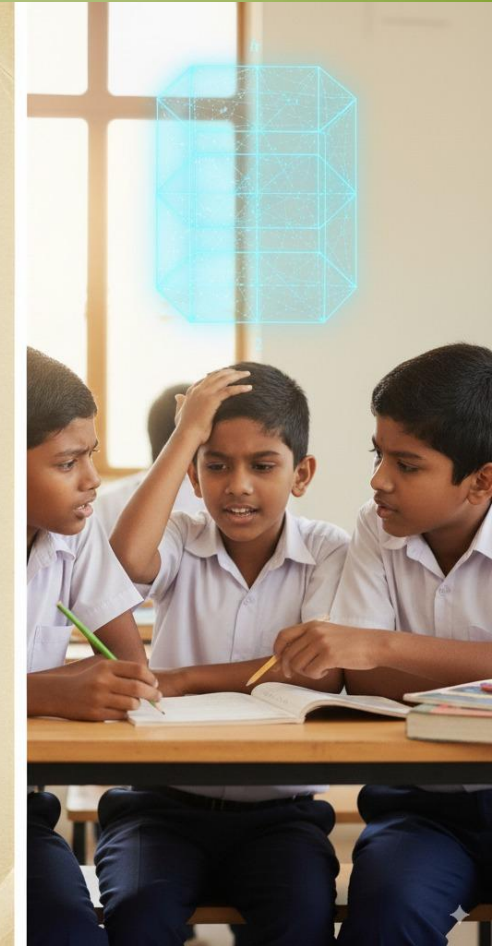
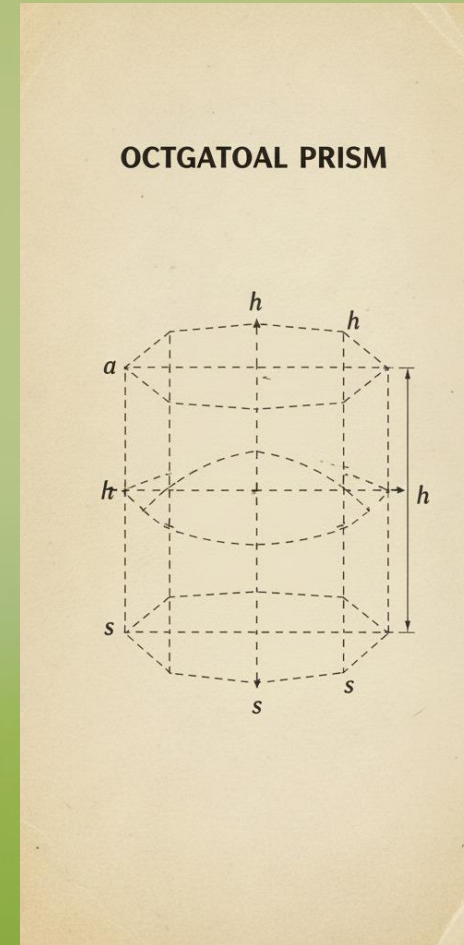
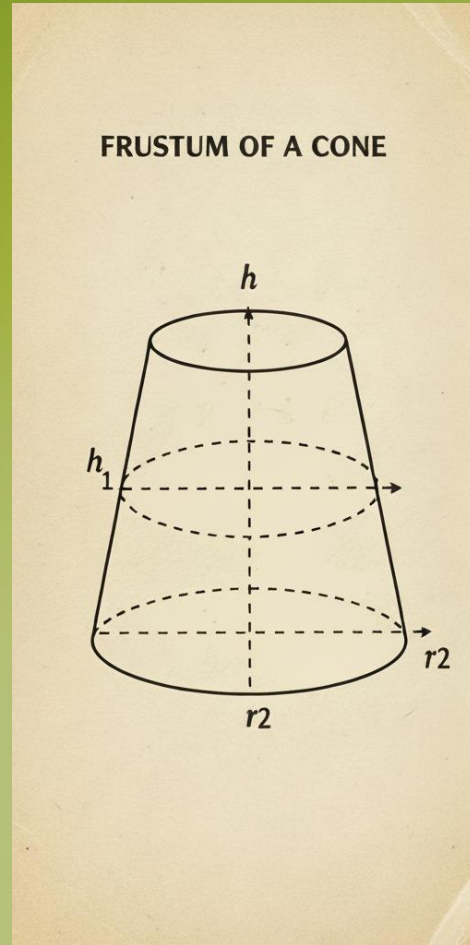
How can an offline, low-cost augmented reality (AR) mobile application, aligned with the sri lankan syllabus curriculum, improve students' conceptual understanding, problem-solving ability, and engagement compared to traditional textbook-based learning?

MATHS LEAD

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300,000 students. Same problem



Knowledge Gap

Maths Lead

No research exists on **AR effectiveness** in Sri Lankan mathematics education with **local curriculum and textbooks** (Sinhala and English)

Proven Solution

AR Mobile App : Transform textbook diagrams into interactive 3D models using proven Unity and marker-based tracking

Existing System Shortcomings

International AR tools **don't work** with Sri Lankan textbooks and **require expensive hardware**

How Our Solution Addresses The Shortcoming

Direct integration with **government textbooks** and optimized for **low-end smartphones (2-3GB RAM)**

Research Study	Identified Research Gap	My Research Contribution (Grade 8 Mathematics - Sri Lanka)
Chonchaiya & Srithamsee (2025) <i>AR for Geometry Learning (Thailand - Vocational School)</i>	Focused on vocational students aged 18-35; limited to basic geometric concepts ; no curriculum alignment with national secondary education standards	Targets Grade 8 secondary students (13-14 years) using official Sri Lankan curriculum alignment with government-issued textbook integration
Bulut & Borromeo Ferri (2023) <i>Systematic Literature Review on AR in Mathematics Education</i>	Identified lack of curriculum-specific AR tools ; most studies use generic AR applications not tied to national textbooks or syllabi	Creates direct textbook-to-AR mapping using Sri Lankan Grade 8 mathematics textbook diagrams as AR markers for curriculum-aligned 3D models
Schutera et al. (2021) <i>cleARmaths (Germany) - Vector Geometry for Grade 12</i>	Focused on advanced vector geometry for Grade 12 ; requires Android devices with higher specifications; limited to German educational context	Targets foundational Grade 8 concepts (basic geometry, algebra, data representation) with preloaded assets optimized for 2-3GB RAM devices , addressing Sri Lankan infrastructure limitations
Elsayed & Al-Najrani (2021) <i>AR for Middle School Mathematics (Saudi Arabia)</i>	Limited to specific chapter (Area & Volume) ; no comprehensive curriculum coverage; no empirical data from South Asian contexts	Covers entire Grade 8 curriculum systematically ; provides first empirical dataset from Sri Lankan classrooms through quasi-experimental study
SHILPA Project (Current Research) <i>B.A.S.Hasith Bulathgama IT22913760 SLIIT</i>	Addresses all identified gaps: Curriculum alignment, low-resource optimization, South Asian validation, user-friendly design for struggling students	Comprehensive solution: Sri Lankan curriculum integration + offline-first architecture + empirical validation + accessible design for resource-constrained environments

System Gap Analysis

Maths Lead

AR Products	Textbook Integration	Offline Functionality	Low-End Device Support	Sri Lankan Curriculum	Grade 8 Specific	Free Access
GeoGebra AR	X	X	X	X	X	✓
MERGE EDU	X	X	X	X	X	X
zSpace	X	X	X	X	X	X
Labster	X	X	X	X	X	X
cleARmaths	✓	X	X	X	X	✓
Proposed System	✓	✓	✓	✓	✓	✓

Specialized Knowledge Domains

Maths Lead



Mathematics Education

Grade 8 Sri Lankan curriculum expertise to map geometry, algebra, and statistics to 3D models

Mobile AR Development

Unity 3D, computer vision, and mobile optimization for real-time 3D rendering

Latest Technologies Used

Unity AR Foundation 5.0

Cross-platform AR development



Vuforia 10.x

Advanced marker tracking system



Expected Success Evaluation

Unity AR Foundation 5.0



Quasi-experimental design involves testing two student groups – one using the AR app and one using traditional textbooks – with identical math tests before and after a 4-week learning

Classroom Learning Enhancement

Instead of drawing static diagrams on the blackboard, they can scan the textbook and show interactive 3D models to the whole class.

Self-Learning for Students

Students in rural or urban areas can revise lessons at home without needing expensive lab equipment.

Bridging Student Gaps

AR helps weaker students who struggle with abstract visualization (like picturing a cube net folding into a cube).

Teacher Training & Support

Teachers can integrate AR into lesson plans as a teaching aid, reducing their reliance on traditional chalk-and-talk methods.

Beyond School – Real-Life Skills

Students trained with interactive visualization develop better spatial reasoning and problem-solving skills.

SYSTEM OVERVIEW & REAL-WORLD APPLICATION

System Components

Input Processing

Camera captures textbook diagrams,
Vuforia recognizes markers, loads
corresponding 3D models

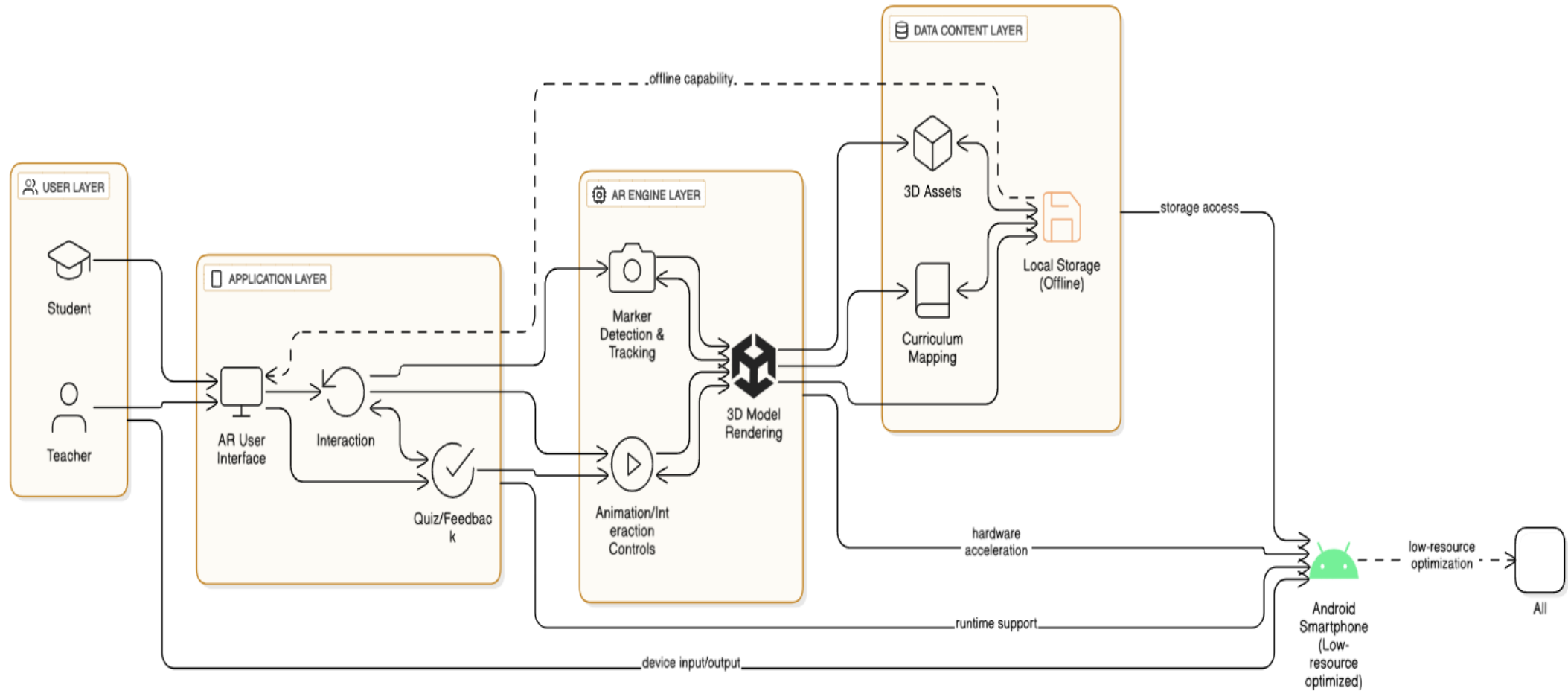


AR Rendering

Unity displays interactive 3D objects
with touch controls for rotation and
manipulation



System Diagram (Single Component)



Maths Lead

Gantt Chart

Maths Lead

Project Gantt Chart

Task		Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8
1.1 Curriculum Analysis	3 weeks	Week 1-3							
1.2 System Design	4 weeks	Week 2-4	Week 1-2						
1.3 Research Methodology	3 weeks		Week 1-3						
2.1 3D Asset Creation	6 weeks			Week 1-4	Week 1-2				
2.2 AR App Development	8 weeks			Week 2-4	All Weeks	All Weeks	Week 1		
2.3 Testing & Optimization	4 weeks					Week 3-4	All Weeks		
3.1 Classroom Implementation	5 weeks							All Weeks	Week 1
3.2 Data Collection & Analysis	3 weeks							Week 3-4	Week 1-2
3.3 Documentation	4 weeks							Week 2-4	All Weeks
Project Completion									▲

Phase 1: Design & Planning

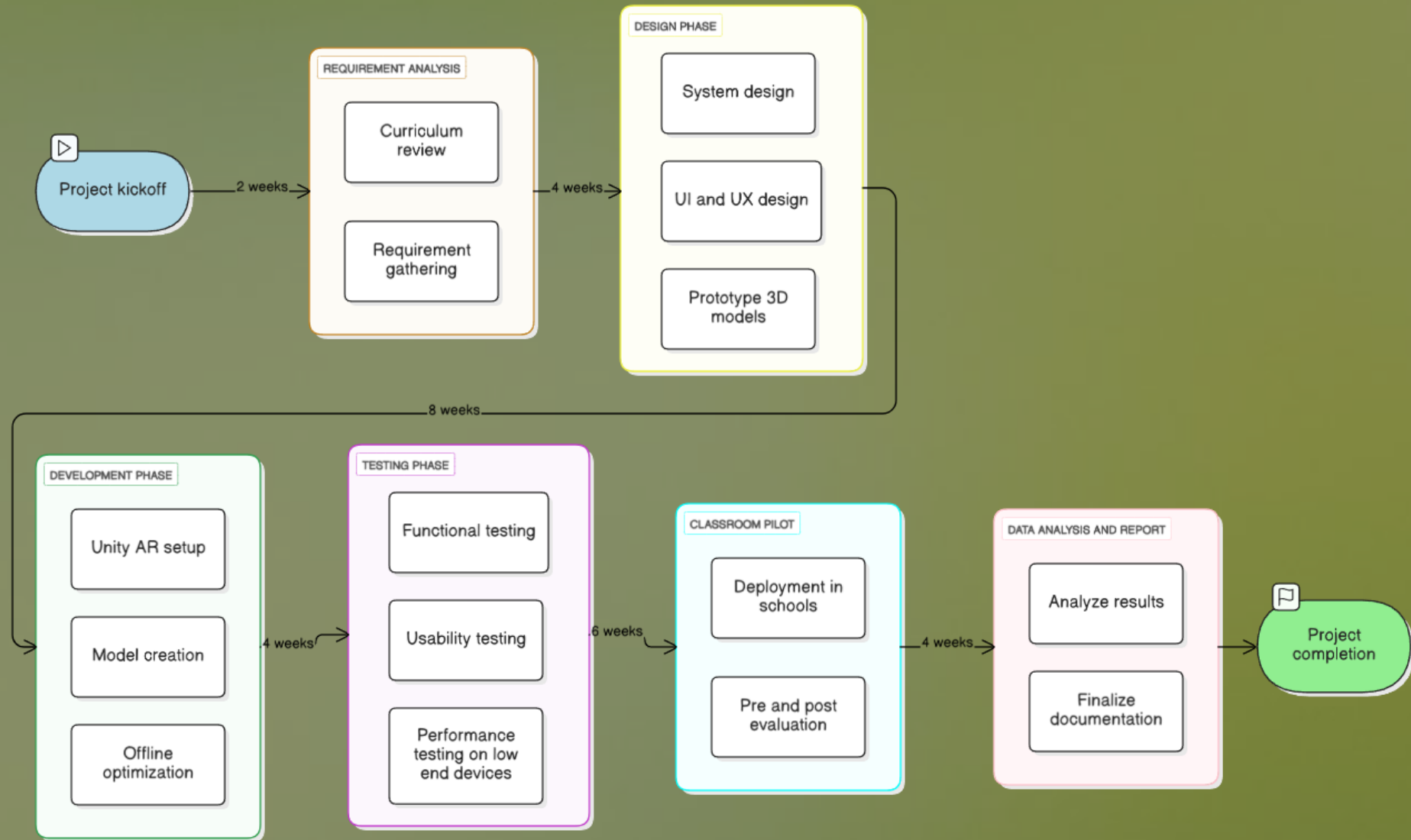
Phase 2: Development & Implementation

Phase 3: Evaluation & Documentation

▲ Milestone

Work Break Down Chart

Maths Lead



- Textbook Diagram Scanning – The app must be able to scan Grade 8 mathematics textbook diagrams and recognize them as AR markers.
- Marker-Based AR Tracking – When a diagram is scanned, the app should display the corresponding 3D model or animation on the device screen.
- Interactive 3D Models – Students must be able to rotate, unfold, and slice 3D shapes (e.g., cubes, cones) and manipulate variables in algebra graphs.
- Offline Functionality – All assets (3D models, animations) should be preloaded to allow full use without internet connectivity.
- Simple User Interface – The UI must be easy enough for Grade 8 students and teachers to navigate with minimal training.
- Feedback/Practice Module – Provide simple quizzes, exercises, or feedback to reinforce understanding after interaction.

- Performance – The app must run smoothly on low-end Android devices (2–3GB RAM) without lag.
- Scalability – The system should be extendable to support additional subjects or higher grades in the future.
- Usability – The design must be intuitive, requiring minimal training for students and teachers.
- Reliability – The app should function without internet and remain stable in classroom use.
- Affordability – The app must be low-cost or free, ensuring accessibility for students across urban and rural schools.
- Compatibility – The app should work across a range of Android versions and devices.

System Gap References

GeoGebra AR

<https://www.geogebra.org/ar>

MERGE EDU

<https://mergeedu.com/>

zSpace

<https://zspace.com/>

Labster

<https://www.labster.com/>

cleARmaths

<https://www.clearmaths.org/>

Research References

Research A

[1] Mehmet Bulut and Rita Borromeo Ferri, “A systematic literature review on augmented reality in mathematics education,” *European Journal of Science and Mathematics Education*, vol. 11, no. 3, pp. 556–572, Jul. 2023, doi: <https://doi.org/10.30935/scimath/13124>.

Research B

[2] Ratchanikorn Chonchaiya and Nattamon Srithammee, “Augmented Reality as a Tool for Enhancing Geometry Learning and Improving Mathematical Understanding,” *ECTI Transactions on Computer and Information Technology (ECTI-CIT)*, vol. 19, no. 2, pp. 350–363, Apr. 2025, doi: <https://doi.org/10.37936/ecti-cit.2025192.260291>.

Research C

[3] S. Schutera et al., “On the Potential of Augmented Reality for Mathematics Teaching with the Application cleARmaths,” *Education Sciences*, vol. 11, no. 8, p. 368, Jul. 2021, doi: <https://doi.org/10.3390/educsci11080368>.

Research D

[4] S. A. Elsayed and H. I. Al-Najrani, “Effectiveness of the Augmented Reality on Improving the Visual Thinking in Mathematics and Academic Motivation for Middle School Students,” *Eurasia Journal of Mathematics, Science and Technology Education*, vol. 17, no. 8, p. em1991, Jul. 2021, doi: <https://doi.org/10.29333/ejmste/11069>.

PHYSICS LEAD



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Physics Component: Problem & Solution

PROVEN GAP :

Lack of interactive and affordable AR tools for Sri Lankan Grade 7 physics.

PROPOSED SOLUTION :

Unity-based AR app optimized for low-end Android phones with marker recognition of textbook diagrams and interactive 3D simulations. (*motion, refraction, circuits, waves*)

KNOWLEDGE GAP :

Students rely on **written memorization** due to **static 2D diagrams** and **poorly equipped labs**.

SHORTCOMINGS OF EXISTING SYSTEMS :

Foreign AR apps are not aligned with Sri Lankan syllabus; most require high-end devices.

RESEARCH GAP ANALYSIS

Research Name	Research Gap	My Research (Physics Component)
Arymbekov & Turekhanova (2025) – <i>AR in Physics Education: A Tool for Intellectual Learning</i>	Focused on foreign contexts; not aligned to Sri Lankan syllabus; tested on well-equipped schools	Develops a syllabus-specific AR app for Sri Lankan Grade 7 physics, targeting local classroom needs
Rahmat et al. (2023) – <i>Mobile AR in Junior High Physics Learning (Indonesia)</i>	Demonstrated AR effectiveness but only in Indonesian schools; no low-resource adaptation	Implements AR optimized for low-end Android devices in Sri Lanka's under-resourced schools
Prahani et al. (2022) – <i>Trend of VR & AR in Physics Learning 2002–2021</i>	Highlighted global progress but identified lack of localized implementations in developing nations	Provides a localized AR solution specifically tailored to Sri Lankan curriculum and context
McDermott & Redish (2009) – <i>Physics Education Research for Middle School</i>	Pointed out persistent conceptual misunderstandings due to reliance on rote memorization and static diagrams	Uses AR to visualize invisible/dynamic physics phenomena (motion, refraction, circuits, waves)

SYSTEM GAP ANALYSIS										
Technical Implementation Comparison - AR Physics Education Systems										
System Features	Unity Engine Support	Vuforia/ARCore Integration	Offline Asset Storage	Low-RAM Optimization	Real-time Physics Engine	Marker Recognition	Cross-Platform Deployment	User Progress Tracking	Teacher Dashboard	Curriculum API
Mobile AR Physics (Rahmat et al.)	~	✓	X	X	✓	✓	X	X	X	X
PhET Simulations (Web-based)	X	X	~	✓	✓	X	✓	~	X	X
Merge EDU (Cube-based AR)	✓	✓	✓	X	~	✓	~	✓	~	X
CoSpaces Edu (Block-based AR)	✓	✓	~	X	~	~	✓	✓	✓	X
Augmented Physics Constructor (Arymbekov & Turekhanova)	~	✓	X	X	✓	✓	X	~	X	X
Google Expeditions (Discontinued)	✓	✓	~	✓	X	✓	✓	X	~	X
Proposed Unity AR System (Hettiarachchi G.V.)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

APPLICATION OF SPECIALIZATION KNOWLEDGE

DOMAINS OF KNOWLEDGE:

Physics Education (conceptual simulations), AR Unity development, mobile computing.

TECHNOLOGIES USED:

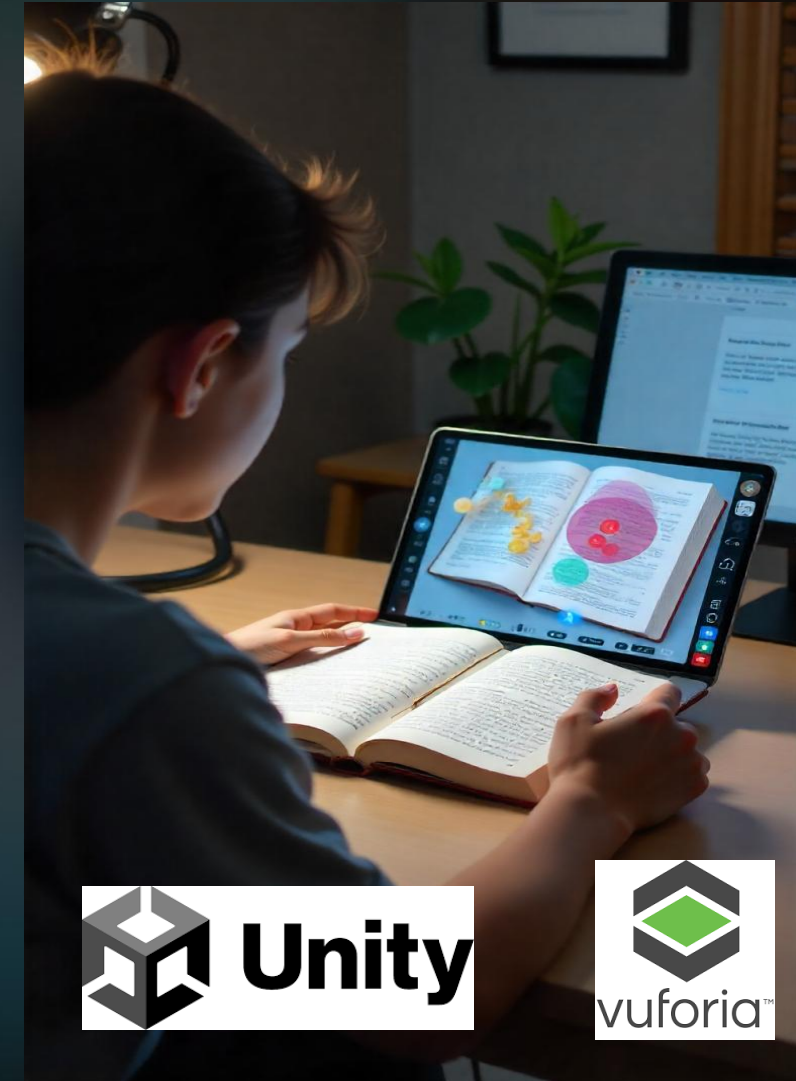
Unity Engine 5.0 (Cross-platform AR development),
Vuforia marker tracking (Advanced marker tracking system)

EVALUATION/VALIDATION:

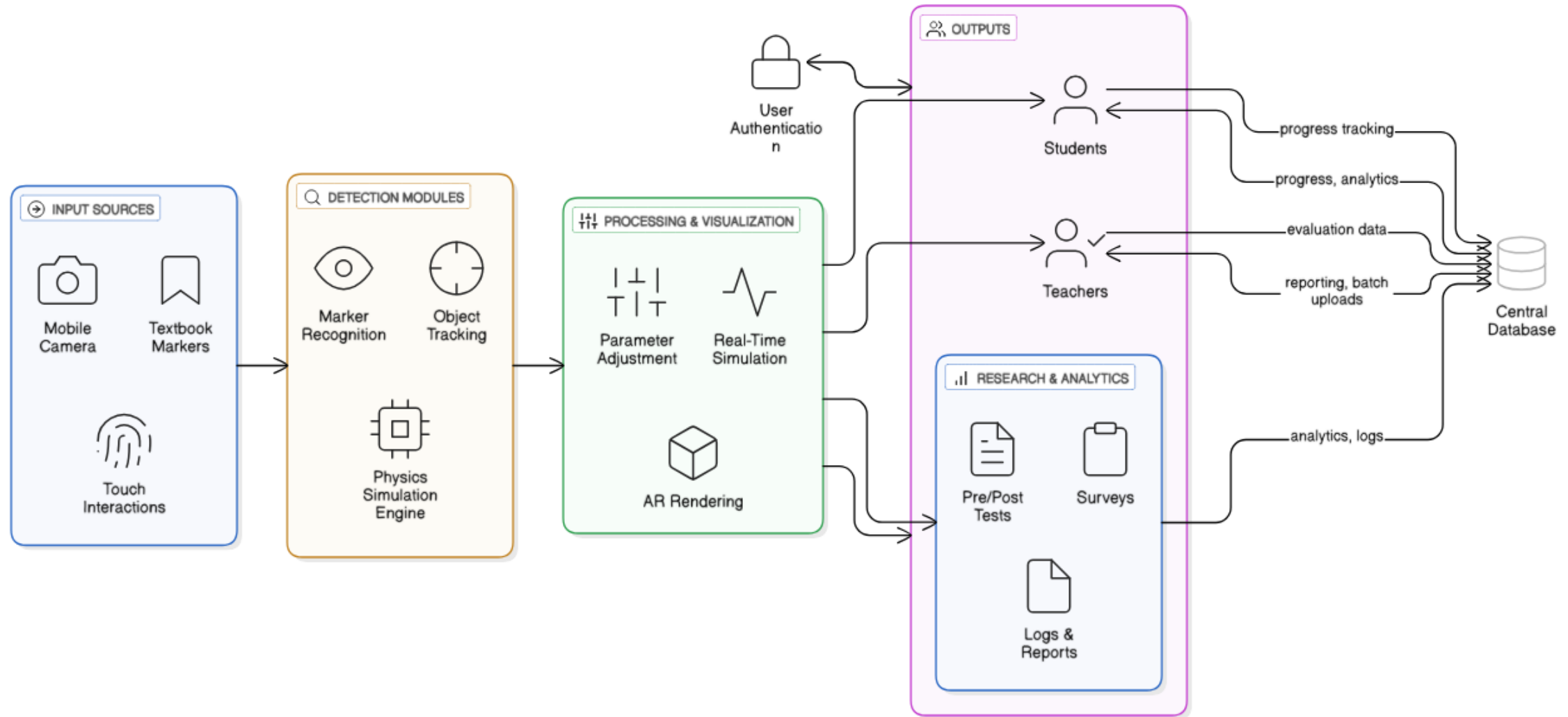
We will test the app using a pre and post-test method to measure learning improvement (normalized gain $\langle g \rangle$), along with surveys and feedback to check usability and student engagement.

DATA & ETHICS:

Grade 7 state-supplied textbooks (markers), school pilot testing, Ministry approval, anonymized surveys/interviews.



INDIVIDUAL SYSTEM OVERVIEW DIAGRAM



IMPLEMENTATION DETAILS

SYSTEM OVERVIEW

Unity AR app **scans textbook diagrams and overlays 3D models to visualize physics concepts.**

REAL-WORLD USE

- Classroom demonstrations
- Student self-study
- Virtual lab substitute
- Examination preparation

Expected Success Evaluation

Unity AR Foundation 5.0 with Physics Simulation Engine: A **quasi-experimental study** will **compare two student groups**; one learning with the AR app and one using traditional textbooks – using identical physics tests before and after a **4-week learning period**

FUNCTIONAL REQUIREMENTS

Marker-based Recognition:

Recognizes Grade 7 physics diagrams (circuits, rays, trajectories) via camera and overlays matching 3D simulations.

Interactive 3D Simulations:

Physics principles shown as manipulable 3D objects (e.g., projectile angle, circuit resistance/voltage).
Real-time variable control with sliders, toggles, and gestures.

Curriculum Alignment:

Simulations directly mapped to Sri Lankan Grade 7 physics syllabus.

Offline Functionality:

All assets preloaded for full offline use.

User Profiles (Future Extension):

Option to save logs (time, simulations viewed) and export data (CSV/Excel) for teachers.

Teacher Support Tools:

Built-in quick guides.
Control to enable/disable specific simulations for lessons.

NON – FUNCTIONAL REQUIREMENTS

Performance:

Runs smoothly on low-end Android (≥ 2 GB RAM, quad-core). Marker recognition ≤ 2 s in normal lighting.

Scalability:

Supports adding future subjects (e.g., Grades 8/9) without recompiling core engine.

Usability:

Intuitive interface for 12–13-year-olds.

Large buttons, simple icons, minimal text, clear instructions.

Reliability & Robustness:

Works in varied classroom conditions (lighting, partial occlusion).

Provides fallback guidance when markers aren't detected.

Portability:

Compatible with Android 8.0+; adaptable for iOS via Unity cross-platform.

Maintainability:

Modular design (AR recognition, simulation engine, UI separated).

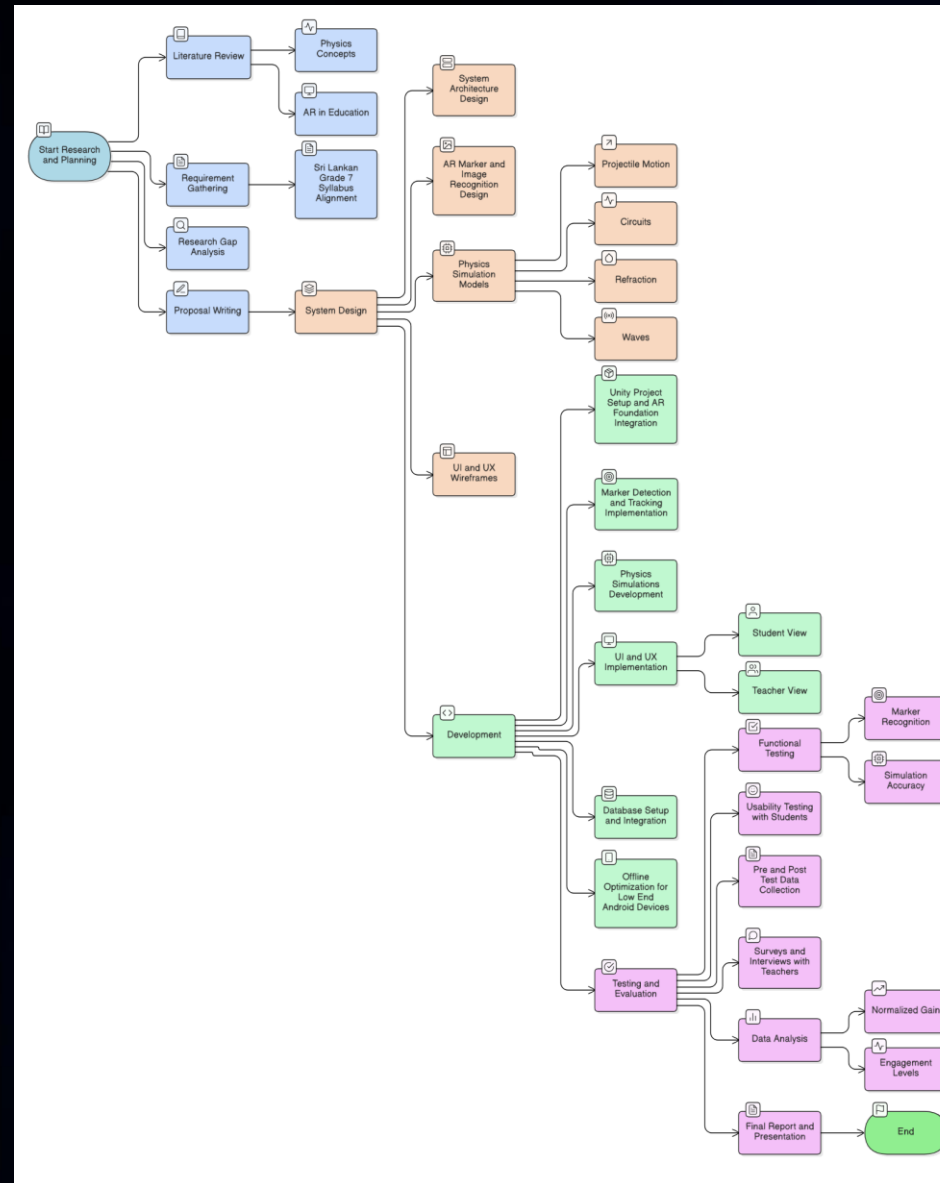
Clear documentation and comments for easy updates.

Security & Privacy:

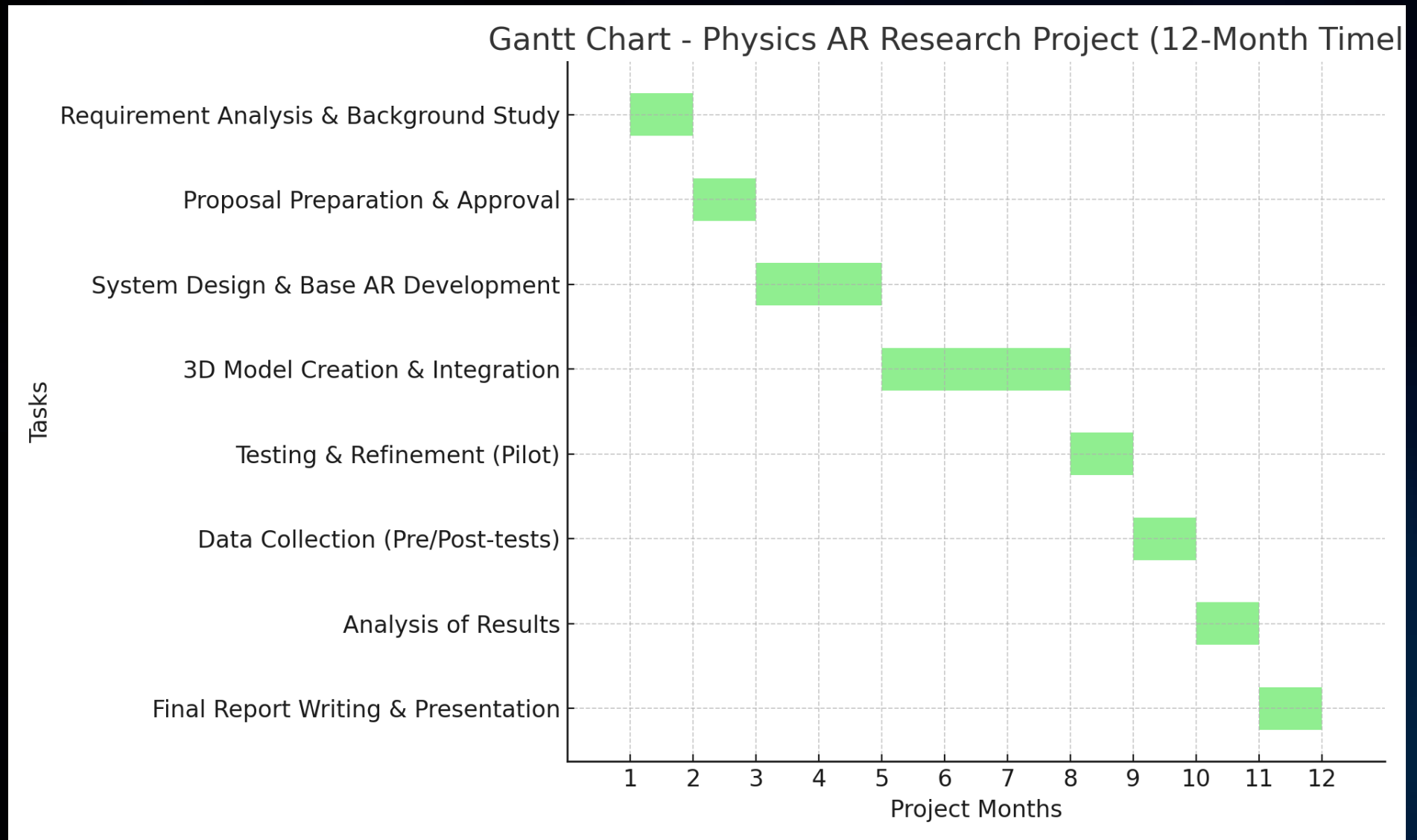
No student data collected/transmitted without consent.

Offline-first design for privacy and low-data use.

WORK BREAKDOWN CHART



GANTT CHART



RESEARCH REFERENCES

- [1] R. Rahmat, A. Datiatur, H. Kuswanto, I. Wilujeng, and R. Perdana, "Implementation of Mobile Augmented Reality on Physics Learning in Junior High School Students," *Journal of Education and E-Learning Research*, vol. 10, no. 2, pp. 132–140, 2023, doi: 10.20448/jeelr.v10i2.4409.
- [2] Z. A. C. Krusberg, "Physics education research: Resources for middle school science teachers," *arXiv preprint arXiv:0903.3183*, 2009. [Online]. Available: <https://arxiv.org/abs/0903.3183>
- [3] B. K. Prahani, I. R. Dawana, and M. A. Kurtuluş, "Trend analysis of augmented reality in science education for supporting SDGs," *unpublished/venue not specified*.
- [4] A. Wibowo, R. Rahmat, H. Kuswanto, I. Wilujeng, and R. Perdana, "Augmented reality in physics education: A tool for intellectual learning," *ResearchGate*, 2024. [Online]. Available: https://www.researchgate.net/publication/389699493_Augmented_reality_in_Physics_education_A_tool_for_intellectual_learning

TECHNOLOGY LEAD

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Proven Gap

No research currently exists on an AR application specifically designed for Sri Lanka's Grade 10 ICT Syllabus.

Proven Solution

AR Mobile App to transform textbook diagrams into interactive 3D models using proven Unity and marker-based tracking

Existing System Shortcomings

AR tools from other countries don't work with Sri Lankan textbooks and need expensive devices.

How Our Solution Addresses The Shortcoming

Works directly with government textbooks and is designed for low-end smartphones (2–3GB RAM).

Research Gap

Research Name	Research Gap	My Research (ICT Component)
Ashabul Taufik et al. (2023) – Development of Augmented Reality-Based Mobile App Learning Media for Computer Network Hardware Material	Limited to single subject (network hardware) in one school; Android-only platform; basic marker-based AR functionality with no curriculum alignment to developing nations	Develops a syllabus-specific AR app for Sri Lankan Grade 10 ICT, targeting local classroom needs with offline functionality
Systematic Review (2020) – Using Technologies as Virtual Environments for Computer Teaching	Found lack of MR applications and integration challenges in educational processes; high costs and resistance to innovation in traditional learning environments	Implements AR optimized for low-end Android devices in Sri Lanka's under-resourced schools with cost-effective solutions
AR Platform (2015) – Visualization and Management Platform with Augmented Reality for Wireless Networks	Focused on network management visualization; required specific hardware (web camera, touch display); no educational context or curriculum integration	Provides a localized AR solution specifically tailored to Sri Lankan Grade 10 ICT curriculum and classroom context
Mixed Reality Study (2016) – Mixed Reality in ICT Networking Education	Required 3D printing infrastructure and physical models; needed structured scaffolding rather than freeform classroom experience	Uses AR to visualize ICT concepts (network topologies, data flow, computer generations) without additional hardware requirements

SYSTEM GAP ANALYSIS

Technical Implementation Comparison - AR ICT Education Systems

EXISTING SYSTEMS	Unity Engine Support	AR Foundation Integration	Offline Asset Storage	Low-RAM Optimization	Real-time Marker Recognition	Textbook Integration	Cross-Platform Deployment	Student Progress Tracking	Teacher Dashboard	Curriculum Alignment
Traditional ICT Textbooks (Current Method)	✗	✗	✓	✓	✗	✓	✗	✗	✗	✓
Taufik et al. AR Network Hardware App (Indonesia)	✓	~	~	✗	✓	✗	✓	~	✗	~
AR Wireless Sensor Visualization Platform (2015)	✓	~	✗	✗	✓	✗	~	✗	✗	✗
Mixed Reality ICT Networking Study (2016)	✓	~	✗	✗	~	✗	✗	~	✗	✗
Generic AR Educational Apps (Global Market)	✓	✓	~	✗	✓	✗	✓	✓	~	✗
Computer Lab Simulations (Sri Lankan Schools)	✗	✗	✓	~	✗	~	✗	✗	✗	✓
Proposed Unity AR System (Sri Lanka Grade 10 ICT)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
✓ Fully Supported ~ Partially Supported ✗ Not Supported										

Knowledge Domains

ICT Education

Grade 10 Sri Lankan curriculum
expertise to map network
topologies, logic gates to 3D
models

Mobile AR Development

Unity 3D, computer vision, and
mobile optimization for real-time
3D rendering

Latest Technologies Used

- Unity AR Foundation 5.0
Cross-platform development for Android
- Vuforia 10.x
Advance marker tracking system



Expected Success Evaluation

Unity AR Foundation 5.0 → A quasi-experimental study will **compare two student groups** , one learning with the AR app and one using traditional textbooks – using identical ICT tests before and after a **4-week learning period**

Data & Ethics

Using Grade 10 government textbooks , testing in schools, Ministry approval, and anonymous surveys or interviews.

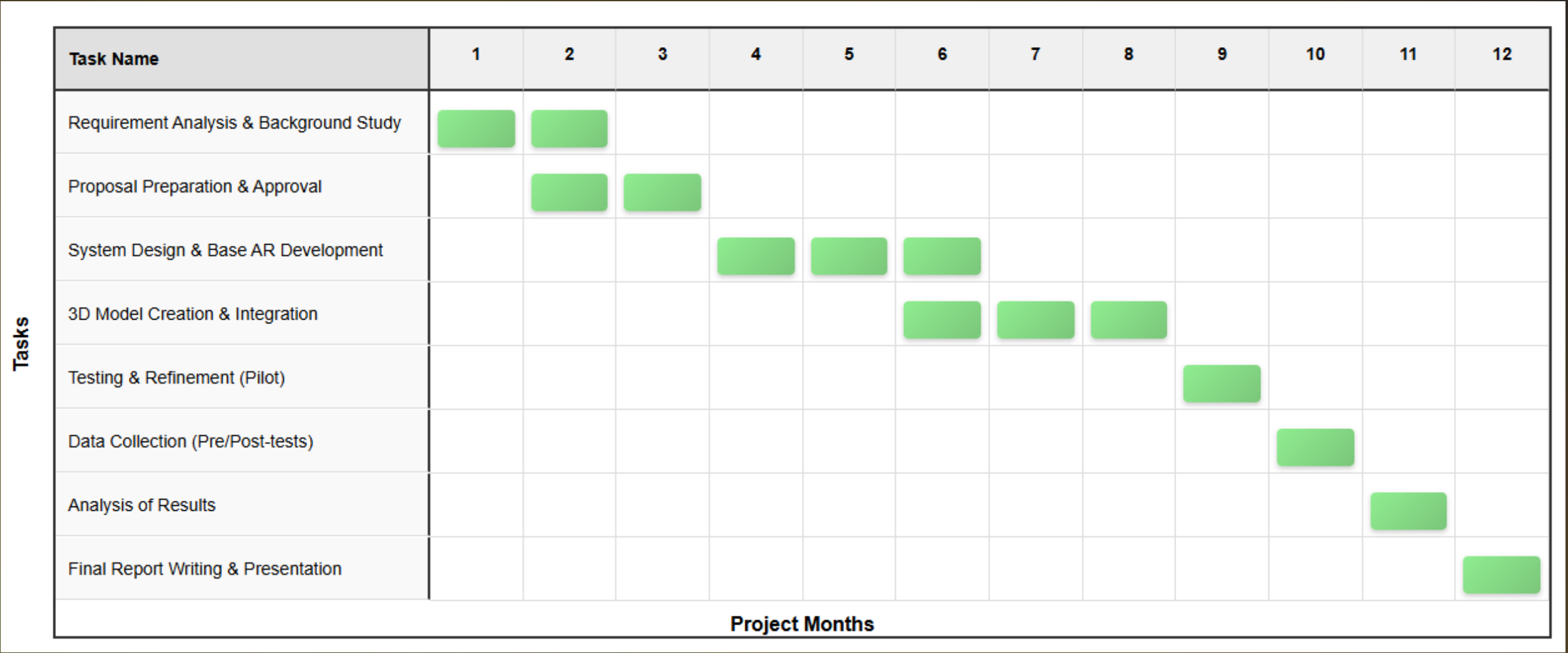
System Overview

- Students can scan textbook diagrams with their phone to instantly see and interact with 3D models on the screen.

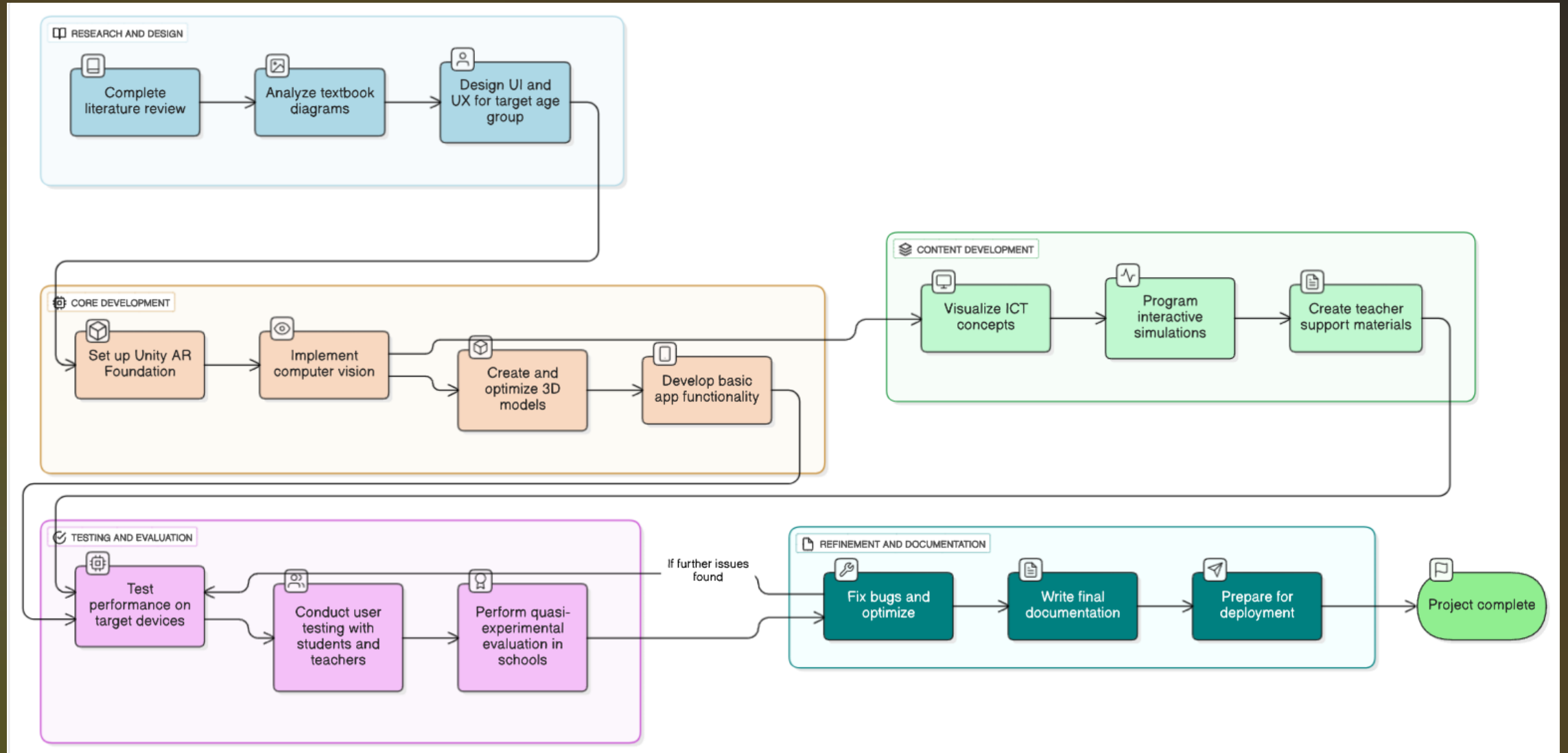
Real-World Use

- Students scan diagrams with their phones during class
- Interactive 3D models help explain complex concepts
- Students can explore at their own pace

Gantt Chart - ICT AR Research Project (12-month timeline)



Work Breakdown Structure



Functional Requirements

- **Marker-based Recognition**

Camera detects ICT diagrams (networks, flowcharts) and overlays 3D visualizations.

- **Interactive 3D Simulations**

Manipulate network data flow, algorithm execution, and database relationships in real-time.

- **Curriculum Alignment**

Mapped to Sri Lankan Grade 10 ICT syllabus (networking, flowcharts, databases).

- **Offline Functionality**

Full offline access with preloaded compilers and simulators.

Non-Functional Requirements

- **Performance**
Works smoothly on basic Android phones (2-3GB RAM) common in Sri Lanka. Recognizes diagrams in under 2 seconds.
- **User-Friendly Design**
Simple interface designed for 15–16-year-olds with clear buttons, minimal text, and easy navigation.
- **Classroom Ready**
Works in different lighting conditions and when diagrams are partially covered. Shows helpful messages like "Adjust camera" when needed.
- **Privacy & Security**
No personal data collection. Works offline-first to protect student privacy and work without internet.

References

[1] K. Sato, N. Sakamoto, and H. Shimada, "Visualization and Management Platform with Augmented Reality for Wireless Sensor Networks," *Wireless Sensor Network*, vol. 07, no. 01, pp. 1–11, 2015, doi: <https://doi.org/10.4236/wsn.2015.71001>.

[1] A. FERNANDO BATISTA, M. THIRY, R. QUEIROZ GONÇALVES, and A. FERNANDES, "Using Technologies as Virtual Environments for Computer Teaching: A Systematic Review," *Informatics in Education*, pp. 201–221, Jun. 2020, doi: <https://doi.org/10.15388/infedu.2020.10>.

Birt, James. "Piloting mixed reality in ICT networking to visualize complex theoretical multi-step problems." (2016).

- [1] A. Taufik, N. Alamsyah, and A. Saputra, "Development of Augmented Reality-Based Mobile App Learning Media for Computer Network Hardware Material," *EduLine: Journal of Education and Learning Innovation*, vol. 3, no. 3, pp. 320–327, Sep. 2023, doi: <https://doi.org/10.35877/454ri.eduline2004>.

ASTRONOMY LEAD

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ASTRONOMY LEAD

Proven Gap

- **Textbooks and diagrams** make astronomy **hard to visualize**.
- Existing AR apps are *global, not curriculum-based, not considered teacher's perspective*.

Solution:

- **Unity based** AR mobile app which **uses marker recognition** to **display interactive 3D models**. Also *quizzes, offline use, and teacher support*.

Knowledge Gap

- **No curriculum-aligned** AR tools.
- **Little focus** on **teachers' perspective** in using AR in classrooms.

ASTRONOMY LEAD

Shortcomings & How Addressed

Current tools

- need internet
- high-end devices
- not syllabus aligned
- focus only on students

Proposed app

- ✓ works **offline**
- ✓ runs on **budget devices**
- ✓ **aligned** with syllabus
- ✓ **teacher-friendly**

ASTRONOMY LEAD

Proven Gap

Research Paper	Key Findings	Identified Gaps
Development of AR-Assisted Teaching Materials in Science Subjects: Solar System Topic (2024)	✓ AR materials improved engagement ✓ Helped students understand solar system concepts	✗ Not curriculum-aligned ✗ Limited interactivity ✗ No teacher perspective
AR Mobile Application to Improve the Astronomy Teaching Learning Process (2022)	✓ AR app increased motivation ✓ Students performed better than traditional methods	✗ Not syllabus-specific ✗ Requires internet ✗ Minimal teacher input
Enhancing Astronomy Skills: Mobile AR in Vietnamese Middle Schools (2024)	✓ AR improved conceptual understanding ✓ Boosted student interest in astronomy	✗ Vietnam-specific context ✗ No offline access ✗ No teacher feedback
Unlocking the Cosmos: Evaluating AR in Secondary Astronomy Instruction (2024)	✓ AR reduced misconceptions ✓ Improved student engagement	✗ Global focus, not localized ✗ Evaluation limited to engagement ✗ No teacher perspective

AR App / Feature	Curriculum Alignment (Sri Lanka Grade 8)	Focus on Astronomy Education	Teacher Dashboard & Role	Offline Access	Language & Local Relevance	Student Evaluation (Quizzes, Tracking)	Research Validation
AR Solar System (Indonesia, 2022)	✗	✓	✗	✗	✗	✓	✓
AR-based Astronomy (Vietnam, 2024)	✗	✓	✗	✗	✗	✓	✓
Unlocking the Cosmos (2024)	✗	✓	✗	✗	✗	✓	✓
AR Science Learning (Philippines, 2021)	✗	✓	✗	✗	✗	✓	✓
AR Astronomy Teaching Material (Turkey, 2020)	✗	✓	✗	✗	✗	✓	✓
Proposed AR App	✓	✓	✓	✓	✓	✓	✓

ASTRONOMY LEAD



Feature / App	Star Walk 2	SkySafari	SkyView	Night Sky	Proposed AR App
Curriculum Alignment (Sri Lanka Grade 8)	✗	✗	✗	✗	✓
Focus on Astronomy Education	✗	✗	✗	✗	✓
Teacher Dashboard & Role	✗	✗	✗	✗	✓
Offline Access	✗	✗	✗	✗	✓
Language & Local Relevance	✗	✗	✗	✗	✓
Student Evaluation (Quizzes, Tracking)	✗	✗	✗	✗	✓
Research Validation	✗	✗	✗	✗	✓

ASTRONOMY LEAD

Application of Specialization-Based Knowledge

Key Domains Applied

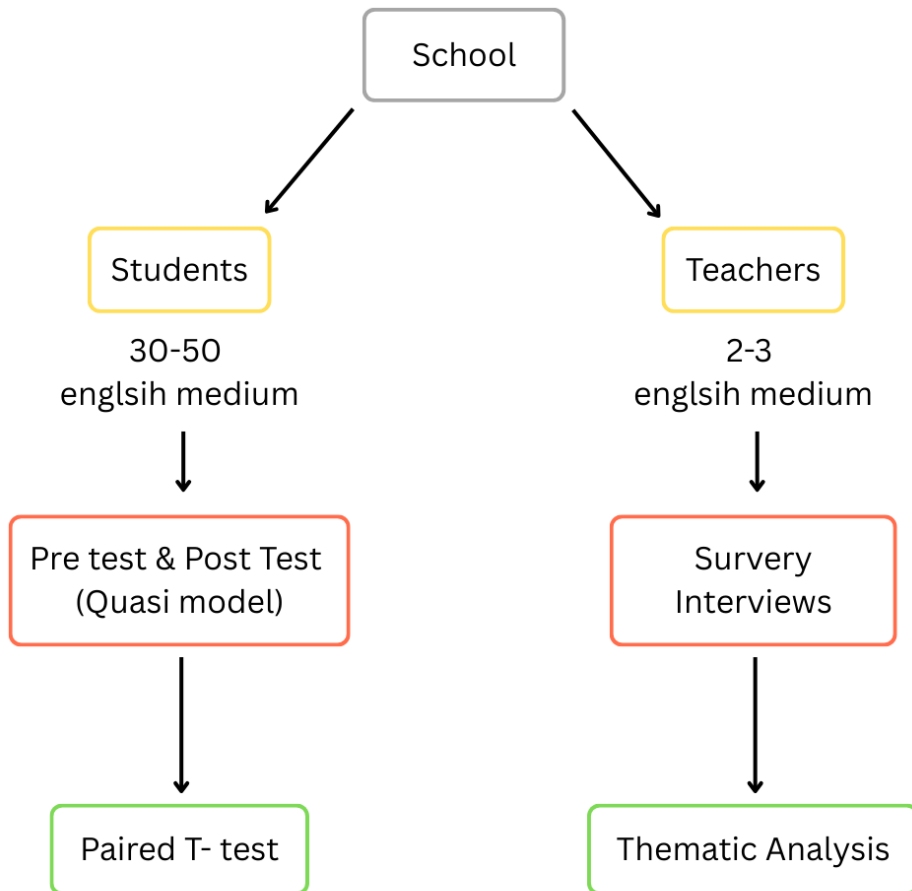
- UI Designing (UI/UX) – designing a **simple, engaging mobile interface**.
- 3D graphics Programming – **immersive 3D models** for astronomy.
- Mobile App Development – **unity 3D engine with AR** foundation.
- Computer Vision – **image recognition** to detect textbook diagrams.

Latest Technologies Used

- Unity + Vuforia → **AR engine** for 3D simulations.
- Optimized 3D Rendering → works on **2-3 gb RAM phones**.
- AR Foundation → **camera access and object tracking**.



ASTRONOMY LEAD



Evaluation / Validation

- Pre-/Post-tests (paired t-test) → **measure learning improvement.**
- Teacher surveys & interviews → **usability and classroom fit.**
- **Expected Outcome** → **higher conceptual understanding & engagement.**

Data & Ethics

- Data from **30–50 Grade 8 students + 2–3 teachers.**
- **Ethical clearance:** parental/teacher consent, anonymous data use, privacy protection.

System Overview

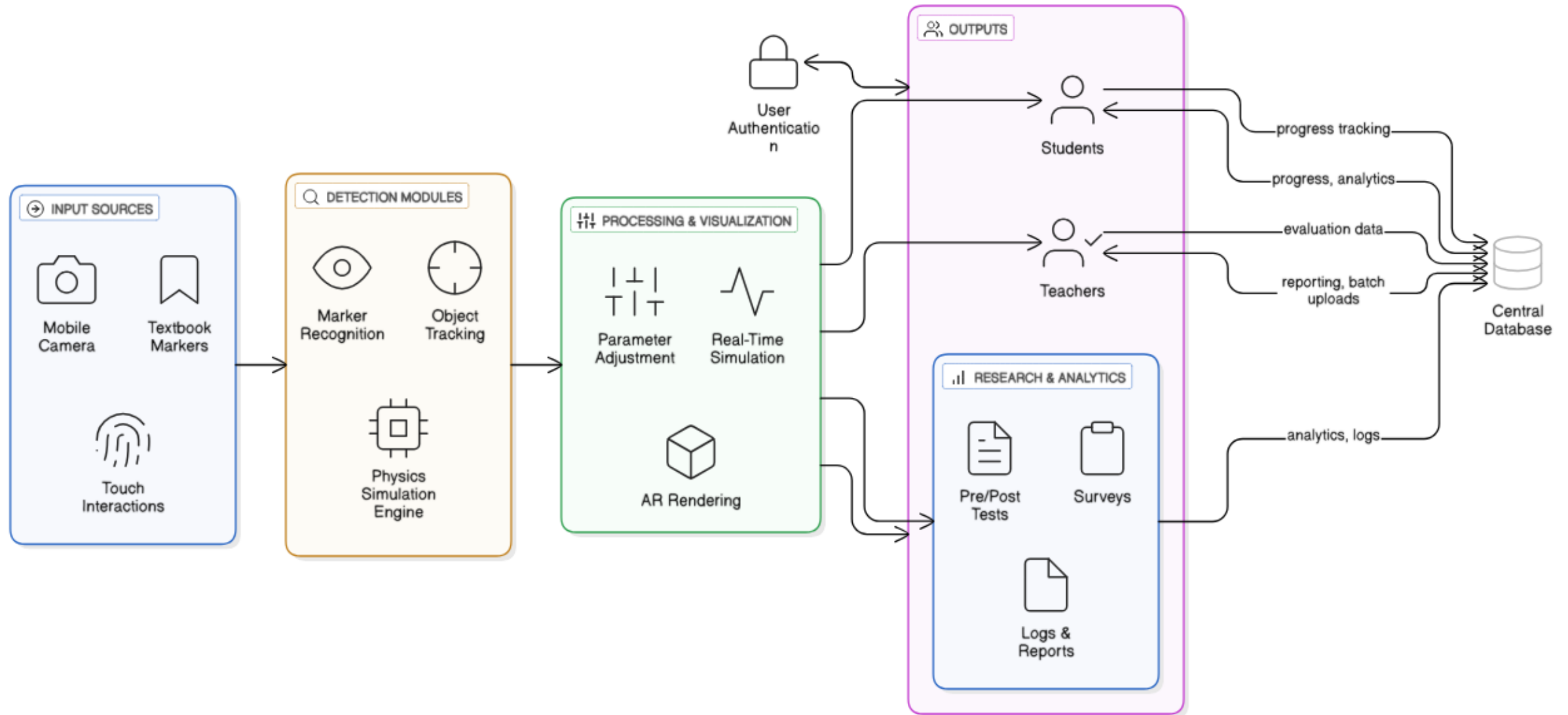
- Students can **scan textbook** diagrams with their mobiles to instantly **see and interact with 3d models** on the screen.

Real-World Use

- Classroom Integration**
Teacher starts the lesson → introduces AR app.
- Explore & Interaction**
Students interact with AR models.
- Assesment through quizzes**
Students test knowledge via quizzes.
- Teacher monitoring & feedback**
Teacher reviews dashboard → adapts teaching.

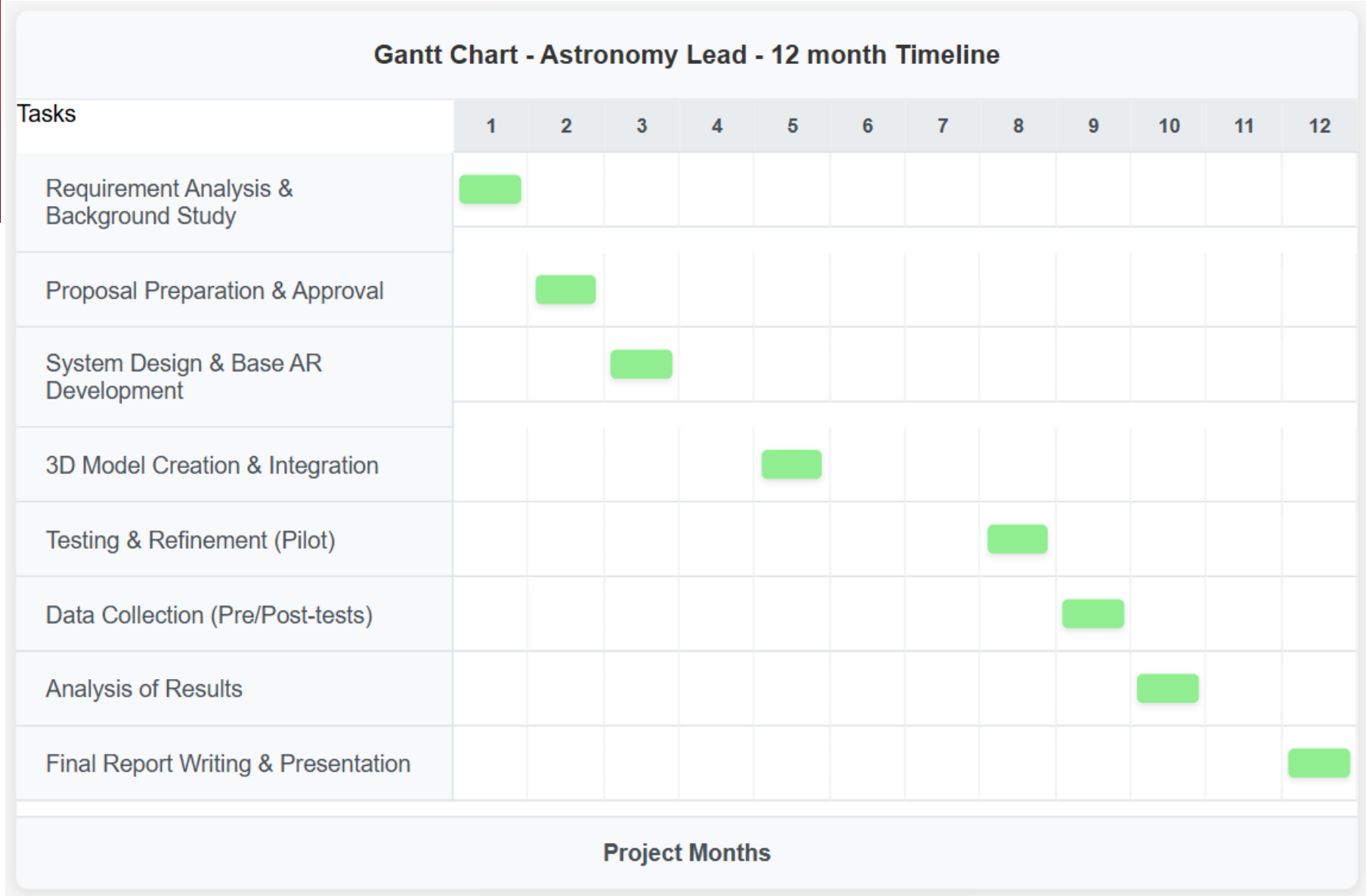
ASTRONOMY LEAD

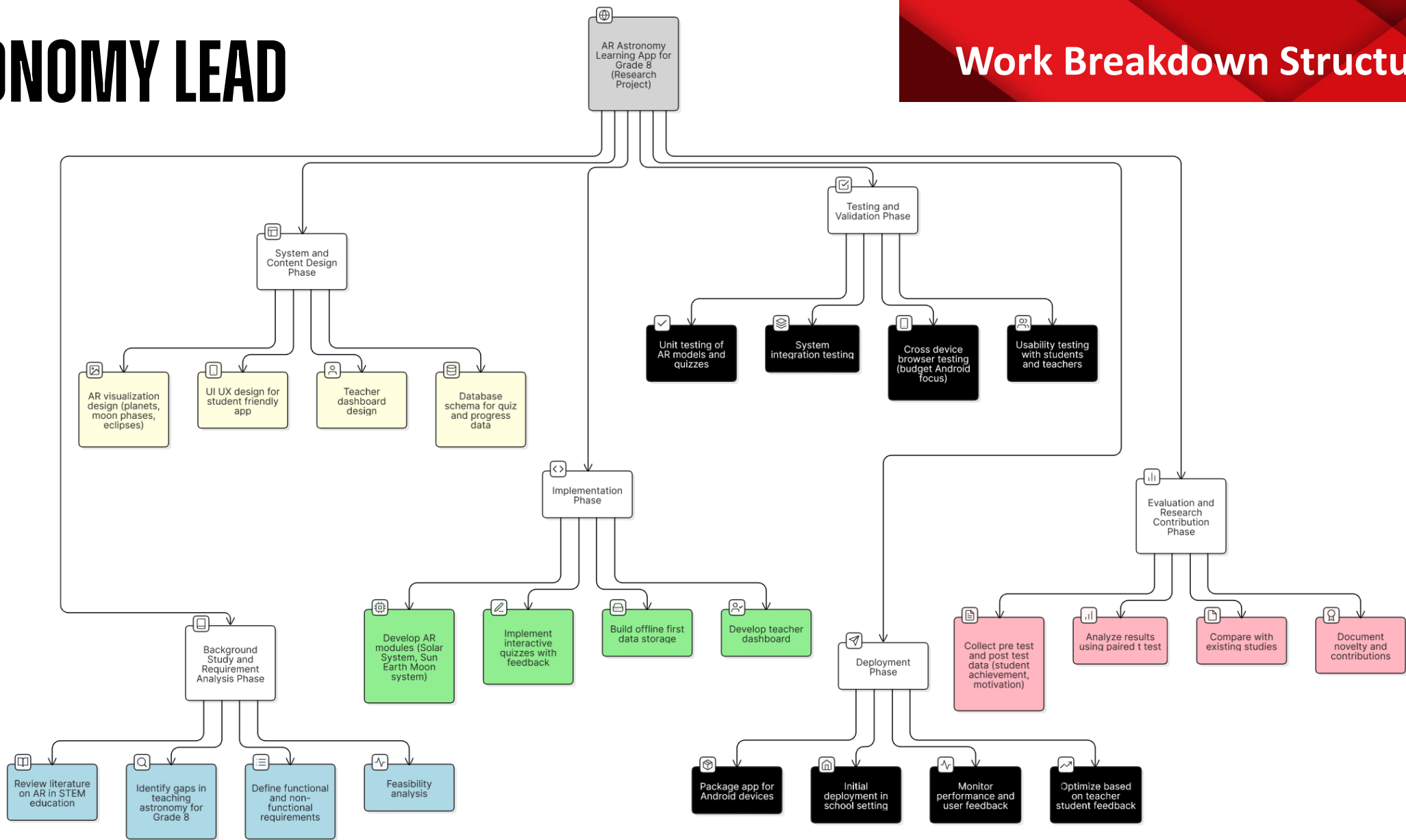
System Diagram



ASTRONOMY LEAD

Gantt Chart





ASTRONOMY LEAD

Functional Requirements

- **Marker-based Recognition** - **Detects diagrams (planets)** through the camera and **overlays matching 3D simulations**.
- **Curriculum Alignment** - Simulations **mapped directly** to the **Sri Lankan Grade 8 astronomy syllabus**.
- **Offline Functionality** - All assets **preloaded** for complete offline access.
- **Gamified Quizzes** - **Instant feedback** with explanations and results.
- **Teacher Tools** - **Quick guides** and controls to enable/disable simulations.

- **Performance** – Smooth on **low-end** Android (≥ 2 GB RAM) **marker recognition** < 2 s in normal lighting.
- **Portability** – Runs on **Android 8.0+** , adaptable to **iOS via Unity**.
- **Scalability** – **Extendable** to new subjects/grades without core recompilation.
- **Maintainability** – **Modular architecture** (AR, simulation, UI) with clear documentation.
- **Usability** – **Child-friendly UI**, large buttons, simple icons, minimal text, clear instructions.
- **Security & Privacy** – Offline-first, **no data collection/transmission** without consent.
- **Reliability & Robustness** – Works under varied classroom conditions, fallback guidance if markers fail.

- [1] Author(s), "AR Solar System: Enhancing Astronomy Education in Indonesia," *Proceedings of the International Conference on Augmented Reality in Education*, Jakarta, Indonesia, 2022.
- [2] Author(s), "AR-based Astronomy: A New Approach to Teaching Astronomy in Vietnam," *Proceedings of the International Conference on Virtual and Augmented Reality in Education*, Hanoi, Vietnam, 2024.
- [3] Author(s), "Unlocking the Cosmos: A Comprehensive Guide to Modern Astronomy," *Journal of Astronomy and Astrophysics*, vol. 58, no. 2, pp. 123-135, 2024.
- [4] Author(s), "AR Science Learning: Revolutionizing Science Education in the Philippines," *Proceedings of the International Conference on Educational Technology*, Manila, Philippines, 2021.
- [5] Author(s), "AR Astronomy Material: Developing Augmented Reality Resources for Astronomy Education in Turkey," *Proceedings of the International Symposium on Educational Technology*, Istanbul, Turkey, 2020.

- Star Walk 2 – [Easily Identify Stars At Night | Mobile Stargazing | Essential Guide To Astronomy | Star Walk](#)
- SkySafari – skysafariastronomy.com/
- SkyView – [SkyView® Lite - Apps on Google Play](#)
- Night Sky – icandiapps.com/night-sky-4/

Investment/Cost (Labor + Material) & Cost Recovery Strategy

Investment Category	Cost Breakdown (LKR)	Pricing & Recovery Model (LKR)
Labor Costs	Development: LKR 2,500,000 Research: LKR 800,000 Testing: LKR 500,000	Freemium Model: Free: Basic Grade 8 content Premium: LKR 900/month full access
Material Costs	3D Assets: LKR 600,000 Software Licenses: LKR 300,000 Hardware/Testing: LKR 400,000	Institutional Licensing: School License: LKR 60,000/year District License: LKR 300,000/year
Marketing & Distribution	Digital Marketing: LKR 800,000 Partnership Development: LKR 500,000	Government Contracts: Ministry of Education: LKR 15,000,000/year Provincial Councils: LKR 4,500,000/year each

Commercialization Roadmap

Year 1: Sri Lankan market penetration (10,000 users) | Year 2: Regional expansion + government partnerships | Year 3: International licensing + advanced subject modules | Long-term: AI-powered personalized learning integration with "Shilpa" brand

Customer Profile

Students: Ages 11–15, struggling with complex STEM concepts and looking for interactive, engaging learning experiences.

Teachers: Seeking affordable and effective tools to help explain abstract concepts and improve student understanding.

Value Proposition

For Students: Turns textbook diagrams into interactive 3D models, making complex concepts easier to understand.

For Teachers: Provides a modern, affordable tool that enhances lesson delivery and student engagement.

Intellectual Property Rights (IPR)

Copyright for the app's content, source code, and 3D models.

Patent opportunities for unique AR technology or educational methods used in the app.

Target Market

Primary Market – Sri Lanka

Secondary Market – Asian Countries (Similar Curriculum)

Main Logo



Secondary Logo



Black and White Logo



Color Palette

